

Biomaterial Playground: Engaging with Bio-based Materiality

Fiona Bell
fiona.bell@colorado.edu
ATLAS Institute
University of Colorado Boulder
USA

Ella McQuaid
ella.mcquaid@colorado.edu
ATLAS Institute
University of Colorado Boulder
USA

Netta Ofer
netta.of@colorado.edu
ATLAS Institute
University of Colorado Boulder
USA

Ethan Frier
epfrier@gmail.com
ATLAS Institute
University of Colorado Boulder
USA

Hyelin Choi
hyelin.choi@colorado.edu
ATLAS Institute
University of Colorado Boulder
USA

Mirela Alistar
mirela.alistar@colorado.edu
ATLAS Institute & Computer Science
University of Colorado Boulder
USA

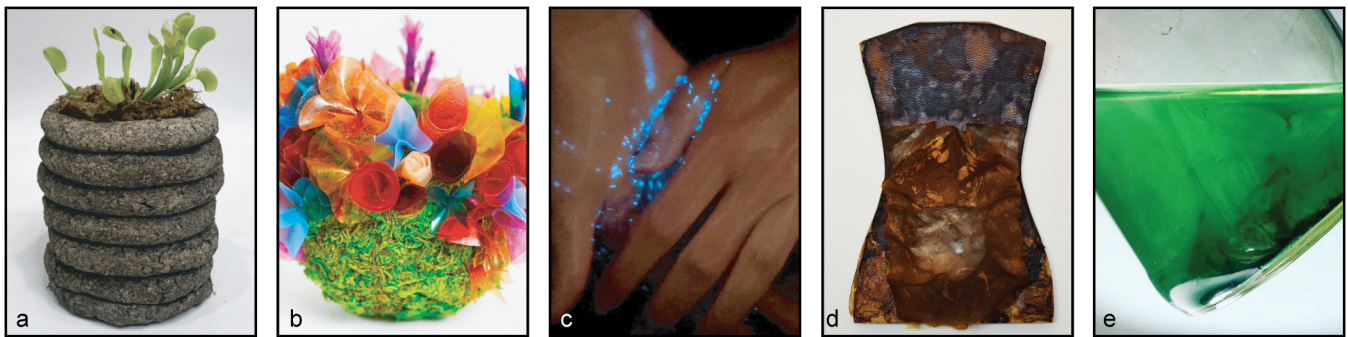


Figure 1: Explorations with Biomaterials: (a) ReClaym, (b) Alganyl, (c) Dinoflagellates (photo by Naharin Shech - used with permission), (d) SCOBY, and (e) Spirulina.

ABSTRACT

Recently, HCI researchers have shown interest in sustainability in the context of both new making methodologies and materials. In this work, we introduce a range of sustainable biomaterials (materials that are bio-based) that we use to create unique interactive interfaces. These biomaterials include ReClaym, a clay-like material made from compost; Alganyl, an algae-based bioplastic; Dinoflagellates, bioluminescent algae; SCOBY, symbiotic cultures of bacteria and yeast; and Spirulina, nutrient-dense blue-green algae. We describe how the implementation of these materials in our designs highlight the importance of utilizing materials that can biodegrade. We also call attention to the importance of care, patience, and understanding during the design processes to facilitate the creation of playful designs that respect the agency of each biomaterial. Lastly, we discuss the gained deeper sense of intimacy and understanding with our biomaterials which not only lead to more personally meaningful interfaces but more sustainable ones.

CCS CONCEPTS

• Human-Computer Interaction → Interface design prototyping.

KEYWORDS

Biomaterials; Bio-HCI; Biodesign; Living Media Interfaces; Sustainability; Materiality; Tangible Interfaces

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1 INTRODUCTION

The waste produced from current making practices plagues digital fabrication and maker culture, resulting in a surge of recent works in HCI that promote the use of biodegradable materials [5, 26, 38], unmaking practices [8, 35, 43], and re-purposing wastes [13, 40]. In this light, we introduce five biomaterials (materials that are bio-based) that can be used to create interactive interfaces and other artifacts. These biomaterials are diverse and cover living organisms as well as processed organisms, including a biodegradable clay made from our personal compost called ReClaym, an algae-based bioplastic called Alganyl [5, 7], bioluminescent algae called Dinoflagellates [27], Symbiotic Cultures of Bacteria and Yeast grown

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in kombucha called SCOBY [1] and a nutrient-dense biomass of cyanobacteria called Spirulina. HCI designers and researchers, including us the authors, have designed interfaces, interactions, and experiences with these biomaterials that highlight the role of discourse on sustainability and more-than-human thinking in HCI.

Our design practice has taught us lessons from engaging with these biomaterials. In this paper, we will unfold our main insights such as: care, patience and understanding and the important roles the play when working with biomaterials. *Care* describes our ethical responsibility to respect and nurture our biomaterials, *patience* describes the empathy required to embrace the slowness of our biomaterials, and *understanding* describes the importance of learning about the limitations and opportunities of each biomaterial. While these values are very important when working with biomaterials in particular, we believe that researchers, designers, and HCI practitioners alike can implement them when working with all materials to create more playful, thoughtful, and sustainable designs. With this Interactivity demonstration, we hope to introduce and promote such biomaterials to the CHI community, encouraging attendees to personally engage with the materiality of biomaterials.

2 RELATED WORK

Materiality in design is becoming increasingly important in HCI [31, 33, 37, 41], with the latest trend suggesting that designers can take inspiration from materials themselves to arrive at a design concept, as opposed to forcing a material to fit a predetermined design concept. With material-centered design approaches, designers seem to be leaning into the “material turn” in HCI [33], in which traditional computers are replaced with computing materials that can be seamlessly integrated into our physical, everyday world [42]. For example, Ishii famously envisioned a future, ‘radical atoms’, in which “all digital information has physical manifestation so that we can interact directly with it” [17]. This concept encourages HCI practitioners to push the traditional boundaries of interfaces by exploring tangible, dynamic materials. In this light, biomaterials (materials of biological origin [29]) act as a unique and exciting spring-board for the conception of new tangible interfaces. Biomaterials have currently been used within the field of HCI to create a variety of interactive interfaces including displays [5, 6, 21, 24, 36], wearable devices [2, 7, 14, 26, 38, 39, 44], art [1, 9, 22, 28], instruments [20], systems [4, 15, 16, 23], games [11, 18, 19, 23, 27], and robots [10, 30]. The use of biomaterials in interface design not only highlights a trend towards sustainability in HCI [8, 25, 32], but also towards more-than-human design goals that consider the interdependencies of humans and other living things [12, 34].

3 BIOMATERIALS WE WORK WITH

The following section presents five biomaterials we commonly work with: ReClaym, Alganyl, Dinoflagellates, SCOBY, and Spirulina.

ReClaym is a clay-like, sustainable biomaterial made from degraded food waste (i.e., compost) that we formulated so it is do-it-yourself (Figure 2.1a). ReClaym is sustainable through the inherent biodegradability and circular life cycle of compost, and personal through the food preferences and habits the material visually and texturally reveals. Because ReClaym often changes based on one’s personal compost (and in turn food/eating habits), it is imperfect

and unpredictable. We support these characteristics of ReClaym by using hands-on making techniques, such as sculpting, molding, and hand-held extruding, that allow the maker to communicate openly with the material. We implemented these techniques to create a range of artifacts that were customized via color, texture, sensing, and conductivity (Figure 2.1b). These artifacts include garden paraphernalia, games, and personal items found in the home (Figure 2.1c). ReClaym’s circular life cycle both begins and ends with composting, thus encouraging sustainable making practices.

Alganyl is a vinyl-like biomaterial cooked from edible ingredients (Figure 2.2a). We began by optimizing a recipe (ratio of ingredients) for Alganyl by characterizing mechanical properties such as strength and flexibility. We then explored how Alganyl can be customized with color, texture, sensing pigments, electrical conductivity, and a variety of textile fabrication techniques (Figure 2.2b). These explorations enabled us to create a collection of Alganyl artifacts which range from do-it-yourself clothing items and environment sensing accessories, to dissolvable food packaging and aesthetic sculptures (Figure 2.1c). Our past work shows that Alganyl can be disposed through composting or fully recycled through re-cooking [5, 7].

Dinoflagellates are microscopic bioluminescent algae that glow when physically agitated (Figure 2.3a). Through our explorations of direct and physical kinetic interactions with Dinoflagellates, we proposed an organism-centered design approach. To this end, we direct designers to take a more-than-human approach, prioritizing the well-being and needs of the organism rather than the human’s. We first focused on designing 3 environments appropriate for Dinoflagellates to thrive in, including: open, porous, and closed environments (Figure 2.3b). We then explored the light characteristics of the Dinoflagellates in each environment and designed playful games in each environment that leverage Dinoflagellates’ interactivity, e.g., checkers, hopscotch, and basketball (Figure 2.3c) [27].

SCOBY is a cellulose-based biofilm that grows at the liquid-air interface in tea (Figure 2.4a). The SCOBY (Symbiotic Culture of Bacteria and Yeast), forms through a complex fermentation process resulting in kombucha. SCOBY grows over the span of 1-4 weeks depending on the kombucha recipe used and environmental factors such as temperature. When removed from their liquid growth medium, the SCOBY dries into a leather-like material, which can be laser cut. We leverage the slow growth of SCOBY to create capacitive sensors (Figure 2.4b) and embed fibers (Figure 2.4c).

Spirulina is the biomass of *Arthrospira platensis* cyanobacteria, a type of blue-green algae (Figure 2.5b). It is highly dense in iron and other nutrients, making it a popular dietary supplement, which comes in a powder form. We grow spirulina in open-source bioreactors that we drain through filters to harvest the living organisms (Figure 2.5a). Once filtered we take use the concentrated Spirulina, which has a gel-like consistency, as a ink for drawing (Figure 2.5c). The Spirulina gel can also be used as a slow light sensor, turning orange as it is exposed to too much light. We can also dry the Spirulina gel to create the more familiar, edible powder.

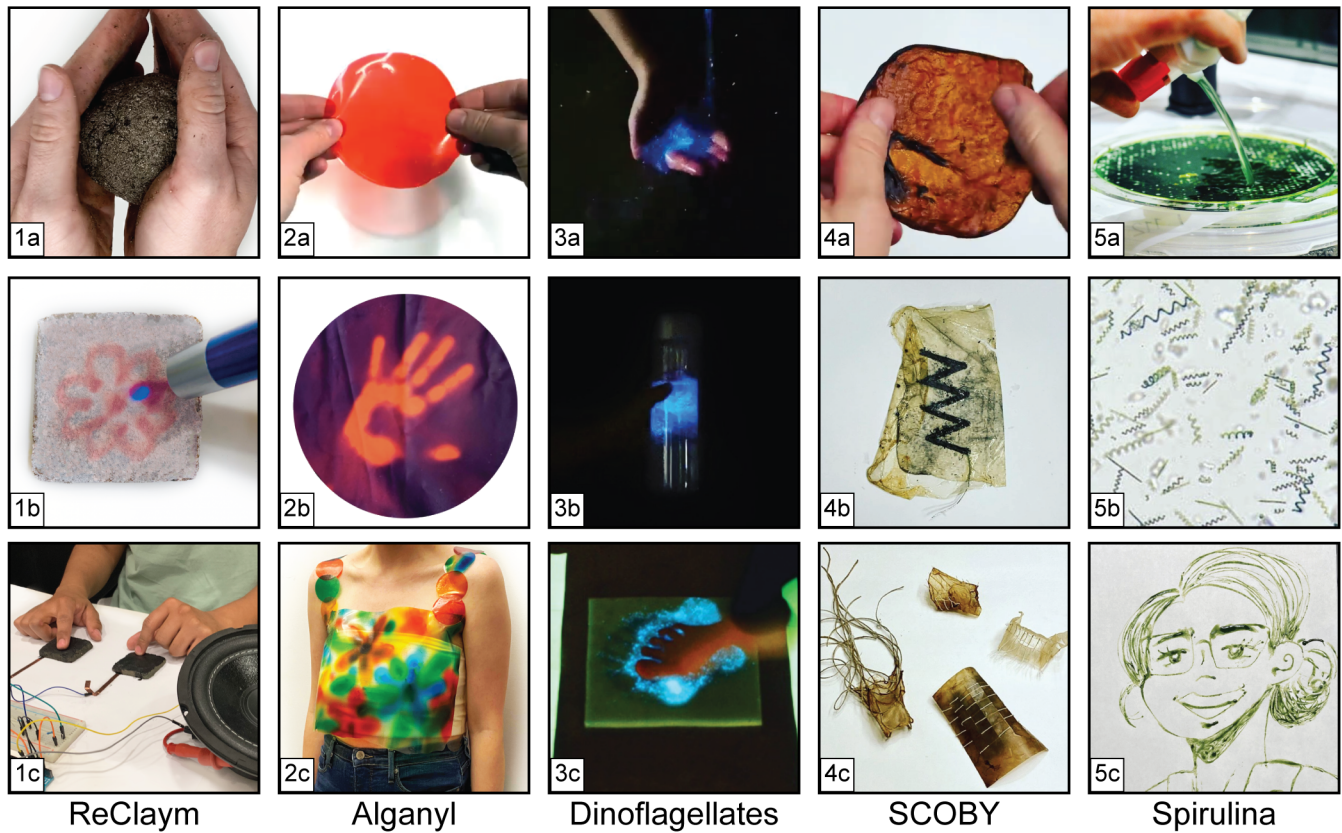


Figure 2: We design with a variety of biomaterials including (1) ReClaym, (2) Alganyl, (3) Dinoflagellates, (4) SCOBY, and (5) Spirulina. We leverage these unique materials to create novel interactions while respecting their agency. Some examples of these interaction include a (1c) ReClaym musical instrument, (2c) Alganyl clothing, (3c) Dinoflagellate hopscotch, (4b) a SCOBY capacitive touch sensor, and (5c) a pen using living Spirulina for drawing.

4 WHAT WE LEARN FROM WORKING WITH BIOMATERIALS

While working with our biomaterials we have gained insights that further guide our design process through a more sustainable approach. This approach includes care, patience, and understanding.

Care is imperative when designing with living materials, as we were required to nurture, assess health, and facilitate appropriate environments for the organism, ultimately taking responsibility for them before, during, and after the design process is completed. We learned that *actions of care are rewarding and allow the maker to form an intimate relationship with the organism*. In our research team's discussions about Dinoflagellates and Spirulina, this aspect of the practice always surprised us since we are well aware of both types of algae being a single-cell, non-sentient organism—yet we could still strongly relate to and have empathy towards them. Through actions of care and physical engagement with these microorganisms, we ultimately developed *kinship*. Although we do not share the same ethical responsibility towards non-living materials, care is still an aspect that surfaces between maker and material. We particularly felt aspects of care and investment come through during the development of Alganyl and ReClaym, which was a

time-consuming process that involved daily iterations of recipes based on our experiences.

Patience is an inherent part of designing and making with biomaterials due to the slowness growth process. However, this slowness often provided us with the much-needed time to meaningfully reflect on past experiences with the biomaterial and consider future design decisions. For example, when working with the SCOBY we had to wait for a week or more to grow our interface. From this, we learned that the slowness of growth *fosters a longer-term relationship between the user and the biomaterial*, and captures unique experiences at a daily pace. Similarly, we experienced slowness when curing Alganyl and when waiting for food to decompose so that we can create ReClaym. In all of the projects, *slowness cultivated a sense of patience, reverence, and respect toward nature, and life*.

Understanding is key to creating interfaces that respect the limits of the biomaterial and reflect the desired design. To gain understanding, the maker and material 'speak' to each other, thus cultivating a deeper and more intimate relationship. This type of communication is facilitated by *real-time feedback loops in which*

the makers' actions enable the material to actively disclose its' capabilities and limitations [3] and the resulting actions of the material provide the maker with sensory feedback. For example, when working with Dinoflagellates, it is essential to shake the cultures daily to manually incorporate CO_2 . While the gas exchange is necessary to maintain the health of the Dinoflagellates, the physical interaction also caused the Dinoflagellates to bioluminesce, allowing us to visually sense their health. Similarly, when designing with ReClaym, we relied mostly on the feel and actions of ReClaym as we physically worked with it to guide the shape and form of our artefacts. Through communication, the maker gains a deeper knowledge of the material, learning it's full range of capabilities and limitations, thus resulting in designs that are best suited for material.

5 CONCLUSION

We present a collection of biomaterials including ReClaym (a biodegradable clay made from personal compost), Alganyl (a vinyl-like bioplastic made from algae), Dinoflagellates (bioluminescent algae that glow when physically agitated), SCOBY (a cellulose-based biofilm grown in tea), and Spirulina (nutrient-dense blue-green algae). Through working with these biomaterials we learned care, patience, and understanding—which lead to designs that are playfully interactive and respectful of each biomaterial's agency. While most of these lessons are inherent to working with biomaterials in particular, we believe they can apply to all designers who create tangible interfaces. Ultimately, by utilizing biomaterials and implementing care, patience, and understanding in one's design practice we hope to create more thoughtful, empathetic, and sustainable designs.

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